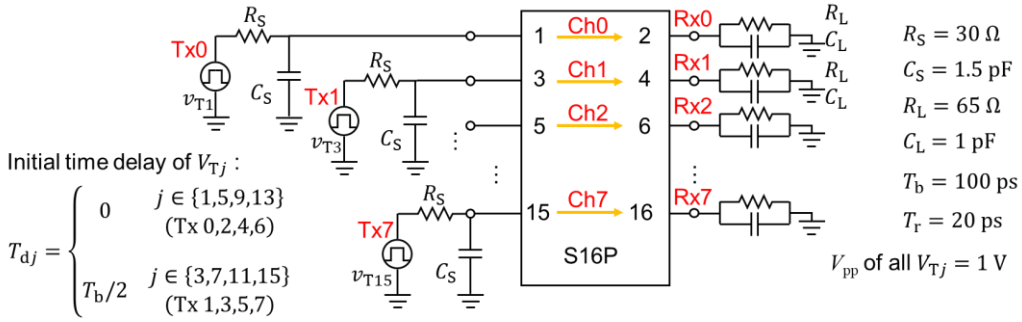
**Homework: Crosstalk simulation**

An S16P is provided to you which is the S parameter of eight channels with coupling, where the input (Tx) ports are 1, 3, 5, 7, 9, 11, 13, and 15, and the corresponding Rx ports are 2, 4, …, 16. Outside the S16P are RC terminations and voltage sources. Your task is to compute the eyes at all Rx's when all Tx's are simultaneously on. All voltage sources generate random and uncorrelated 10 Gbps single-ended NRZ signal with $V_{pp}$ = 1 V and risetime 20 ps. The initial time delays of $v_{T1}$, $v_{T5}$, $v_{T9}$, and $v_{T13}$ are 0 ps, while the time delays of $v_{T3}$, $v_{T7}$, $v_{T11}$, and $v_{T15}$ are all 50 ps (0.5 UI).




In your report, provide the following:

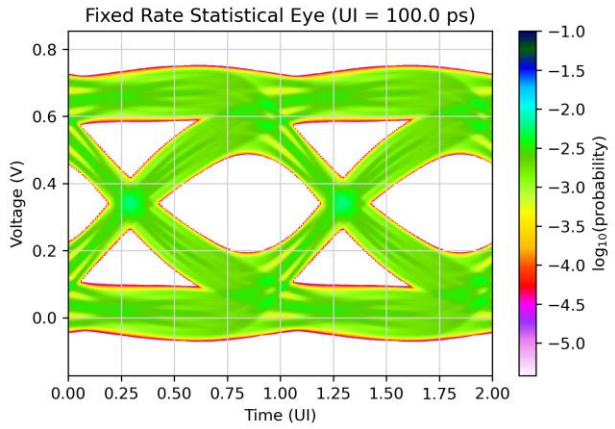
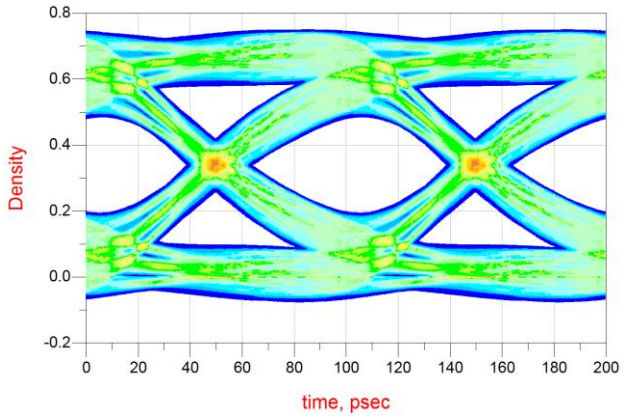
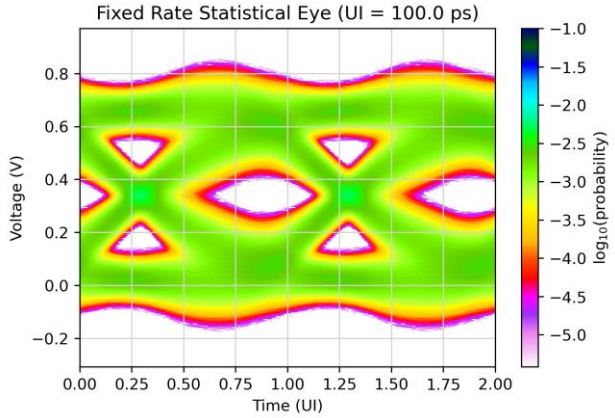
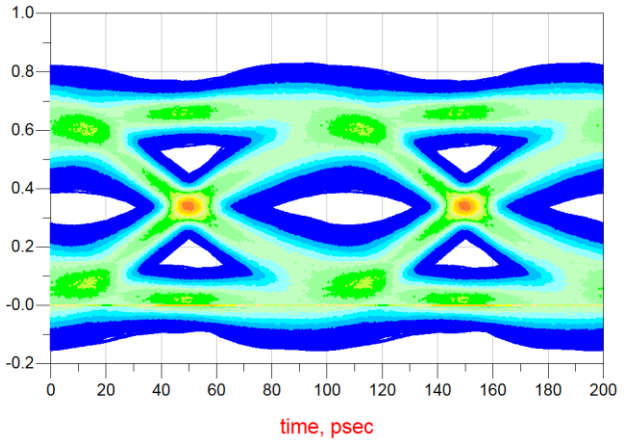
1. A figure showing $S_{11}$, $S_{12}$, $S_{13}$, … $S_{1,16}$.
2. Rx0 eye when only Tx0 is on.
3. Rx0 eye when all Tx's are on.
4. Rx4 eye when only Tx4 is on.
5. Rx4 eye when all Tx's are on.
6. A 4-ns snapshots (time domain waveform) at Rx1~ Rx4 when only Tx4 is on.

Note: although this homework only askes you to show the results at Rx0 and Rx 4, you still need to compute the eyes at all Rx's. In this way, you construct a fast channel simulation tool yourself!

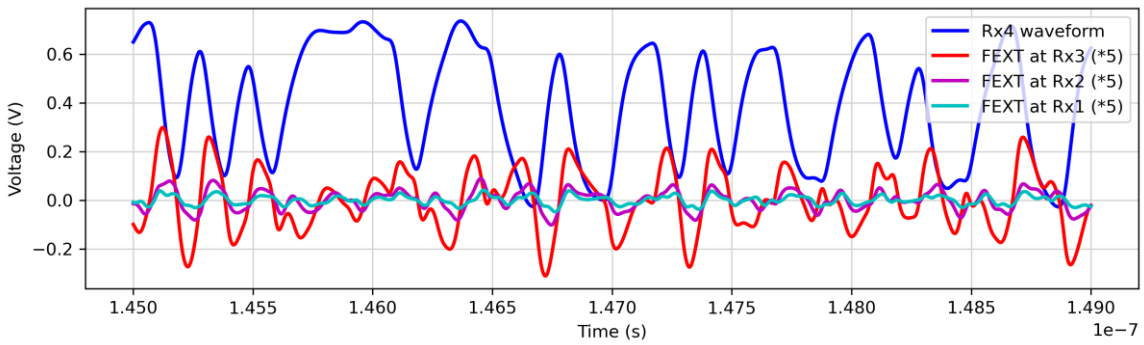
Optional problem: Simulate the circuit in ADS ChannelSim to verify your results.

## Appendix 1: Example answer

Nbp = $2^{18}$

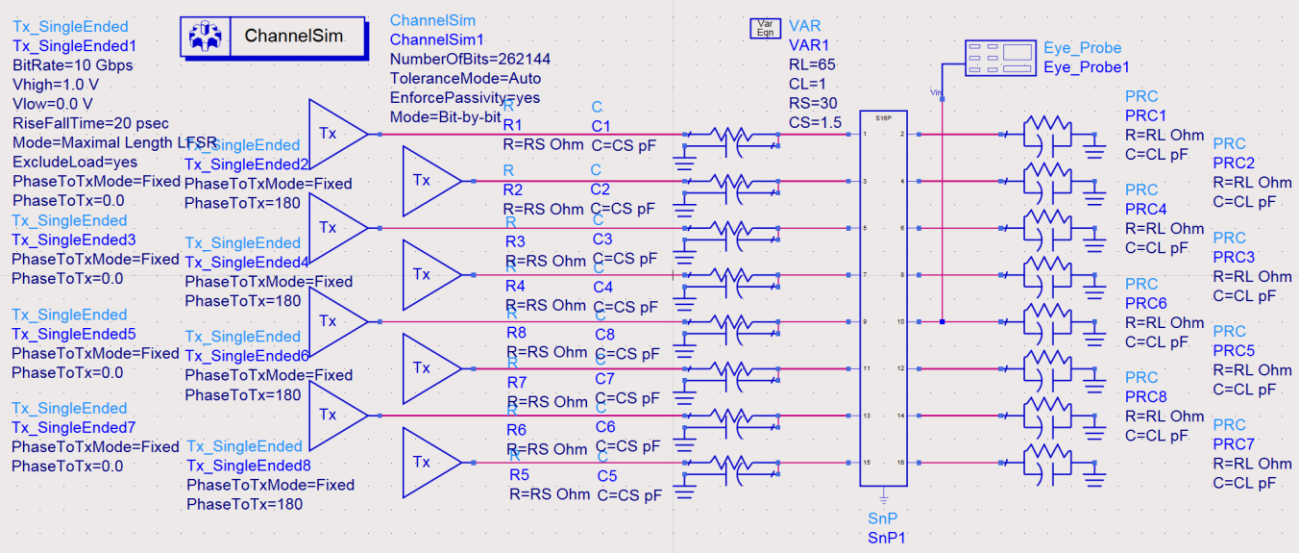
| Python's eyes | ADS's eyes |
|---|---|
| Rx4 eye without crosstalk (only Tx4 is on) | |
|  |  |
| Rx4 eye with crosstalk (all Tx on) | |
|  |  |

Waveforms (only Tx4 is on). FEXT waveforms are enlarged by 5 times to make it easier to examine.



## Appendix 2: ADS setting

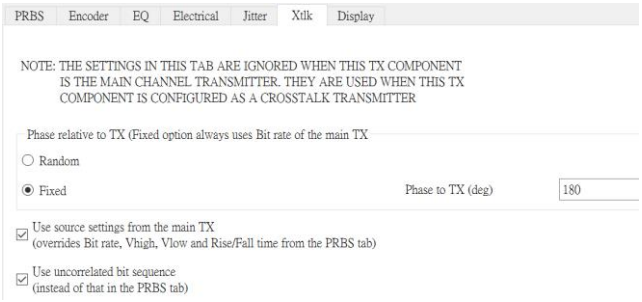
To simulate the eyes in ADS channelsim, connect the circuit as below.



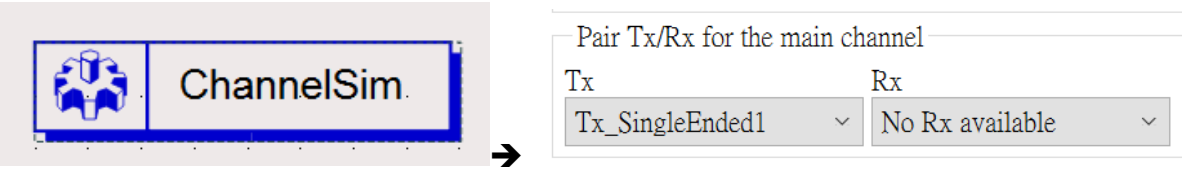

(1) Be careful of port ordering of SNP: 1357… at left, 2468… at right.

(2) Remember to set # taps (register length) to 16 or larger.

(3) In the Xtlk page of the Tx setting, set the "PhaseToTxMode=Fixed" and set "PhaseToTx" properly. 180 deg means 0.5 UI.




(4) In the ChannelSim setting, set the Tx that you want it to be the main channel. For this homework, you should choose the Tx's that correspond to Tx0 and Tx4, one at a time. This is an important step in ADS; if not specified, the results would be wrong.




In my computer, ADS takes nearly 2 min to compute one case. Since there are eight channels, it would take > 10 min to compute the eyes at all Rx's. Hence, the efficiency of sectional convolution is clear.